

# JAYESH SHARMA

Software Engineer | Systems Programming | Full Stack Development

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## EDUCATION

**Government Engineering College Ajmer** — Bikaner Technical University

*Expected July 2027*

Bachelor of Technology in Computer Science and Engineering • CGPA: 7.15 / 10.0

**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Computer Networks, System Design

## TECHNICAL SKILLS

**Languages:** C++, C, Python, JavaScript (ES6+), TypeScript, SQL, Bash

**Frontend:** React.js, TypeScript, Redux, Zustand, TailwindCSS, HTML5, CSS3, Responsive Design

**Backend & Systems:** Node.js, Express.js, Socket.IO, REST APIs, Flask, Unix System Calls, Process Management

**Databases:** MongoDB, MySQL, SQLite, Mongoose ODM, Database Design, Query Optimization

**Developer Tools:** Git, GitHub, VS Code, Postman, GDB, Makefile, Linux/Unix CLI, Docker (basic)

**Computer Science Fundamentals:** Data Structures, Algorithms, Object-Oriented Design, Operating Systems Concepts, Database Systems, Computer Networks, System Design Principles

## TECHNICAL PROJECTS

### Custom C++ Compiler with Multi-Phase Analysis

*C++, Flex, Bison*

- Architected and implemented a multi-phase compiler from scratch supporting lexical analysis, recursive-descent parsing, semantic checking, and intermediate code generation for a custom language grammar.
- Engineered a symbol table manager with scope-aware variable tracking and type checking, reducing semantic errors by providing context-aware validation across nested scopes.
- Designed a modular error-handling pipeline that maps tokens to source locations, pinpointing syntax and semantic errors with line/column precision — reducing average debugging time in test scenarios.

### Unix-Like Shell with Process Management

*C, Linux System Calls, Makefile*

- Built a Unix shell from the ground up using low-level Linux system calls (fork, execvp, waitpid), implementing process creation, execution, and synchronization primitives.
- Implemented built-in commands (cd, help, exit) with custom command parsing, tokenization, and environment variable handling to replicate core shell functionality.
- Demonstrated practical knowledge of operating system concepts: process lifecycle management, system call interfaces, and inter-process communication mechanisms.

### Interactive Algorithm Visualization Platform

*React, TypeScript, TailwindCSS*

- Developed a real-time graph algorithm visualizer rendering BFS, DFS, and Dijkstra's shortest-path traversal with step-by-step animation, speed control, and interactive grid manipulation.
- Architected reusable custom React hooks for grid state management and animation sequencing, achieving clean component separation and enabling future algorithm extensions with zero refactoring overhead.

### Sudoku Solver with Backtracking Visualization | *React, Recursive Algorithms, TailwindCSS*

- Created an interactive Sudoku solver visualizing the backtracking algorithm in real-time with configurable animation speed and step-by-step recursion path highlighting.
- Designed a modular React component architecture supporting user puzzle input, constraint validation, and dynamic UI updates for active backtracking states.

## ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Solved 200+ data structures and algorithms problems across LeetCode (Top 8.6%), GeeksforGeeks, and CodeStudio with emphasis on arrays, trees, graphs, dynamic programming, and greedy algorithms.

**GitHub Portfolio:** [github.com/574-jayeshSh](https://github.com/574-jayeshSh) • All projects available with source code, documentation, and setup instructions